# Demo - Sandbox

Expérimentations

PAC

Ludovic Charleux
ludovic.charleux@univ-smb.fr
Polytech Annecy Chambéry

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# 1 Calculs sur les vecteurs

On peut afficher des vecteus formels:

$$\begin{bmatrix} \theta \\ \alpha \\ 0 \end{bmatrix}_0$$

On peut calculer des choses avec:

$$a_0 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}_0$$

Et:

$$a_1 = \begin{bmatrix} 6 \\ 5 \\ 4 \end{bmatrix}_0$$

On peut additionner les vecteurs:

$$a_0 + a_1 = \begin{bmatrix} 7 \\ 6 \\ 4 \end{bmatrix}_0$$

Et les soustraire:

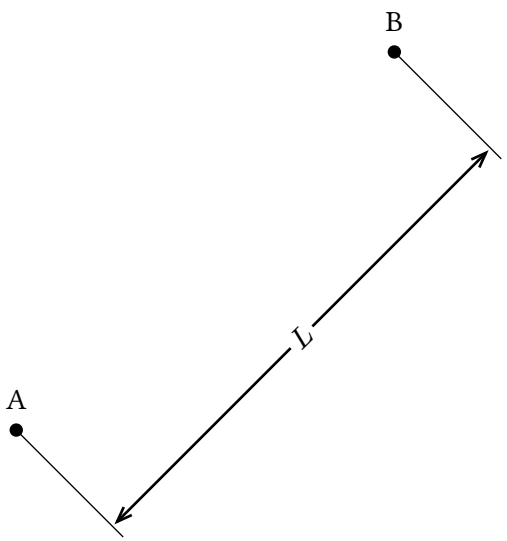
$$a_0 - a_1 = \begin{bmatrix} -5 \\ -4 \\ -4 \end{bmatrix}_0$$

Ou normaliser le vecteur $\vec{a_0}$ en $\vec{v_0}$:

$$\vec{v} = \begin{bmatrix} 0.7071067811865475 \\ 0.7071067811865475 \\ 0 \end{bmatrix}_0$$

```
(10, 10, 0)
```

# 2 Dessins avancés



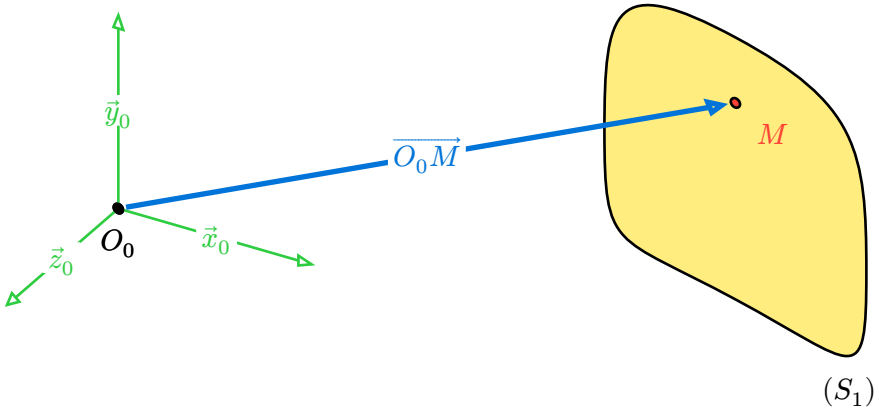

# 3 Gregoire

# 4 Classes

tutu

123

# 5 Repères en 3D



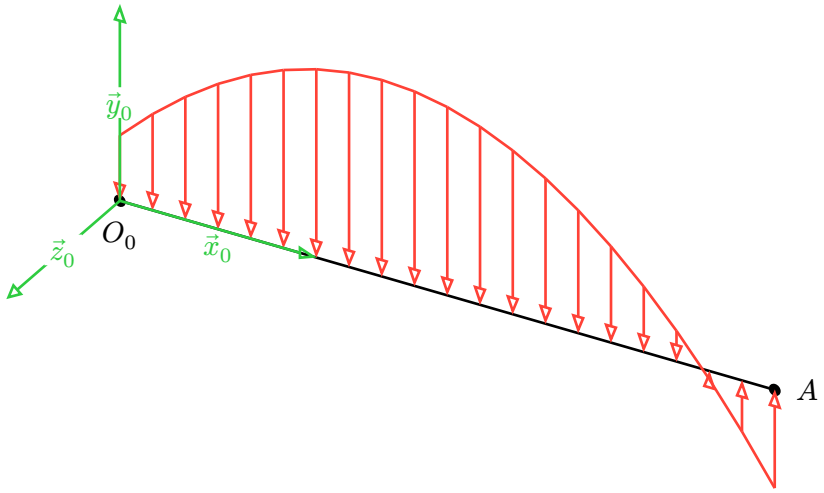

# 6 Torseurs

$$\left\{\begin{matrix} F_y & 0 \\ 0 & 0 \\ 0 & LF_y \end{matrix}\right\}_{O_0,\mathcal{B}_0}$$

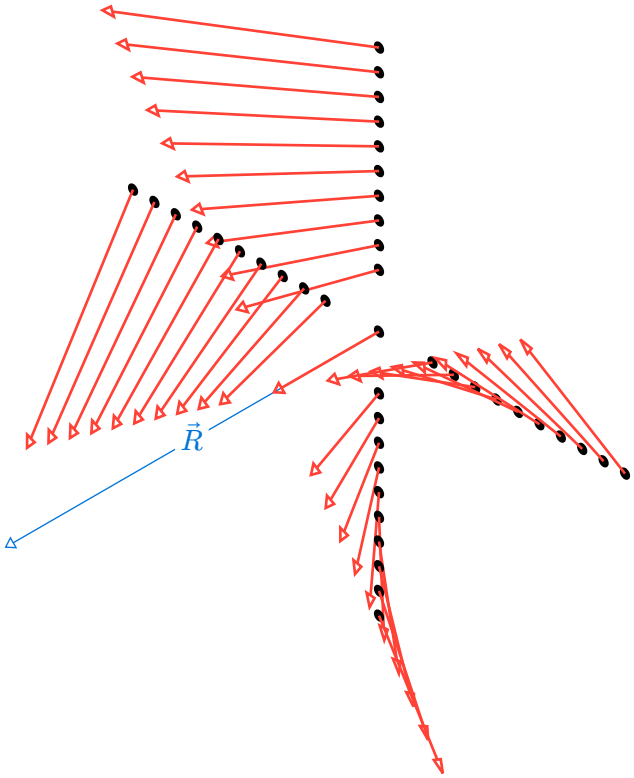
$$\left\{\begin{matrix} \vec{F}_y \\ \vec{0} \end{matrix}\right\}_A$$

$$\{T_0\}$$

# 7 RDM

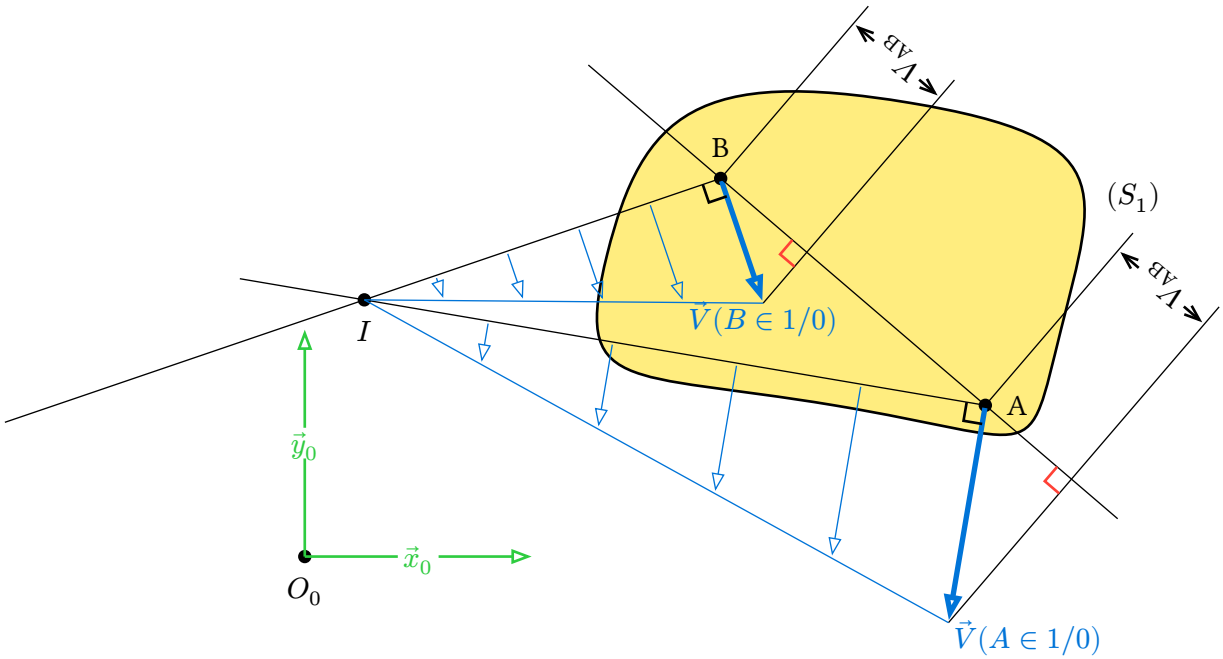



# 8 Champ de moment d'un torseur

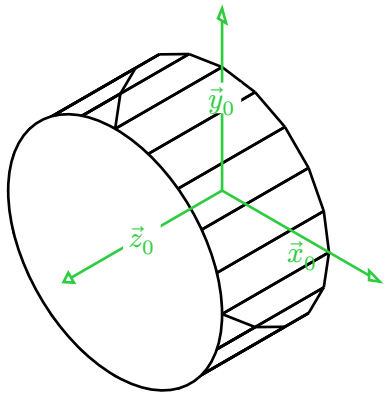



## 9 Champ équiprojectif

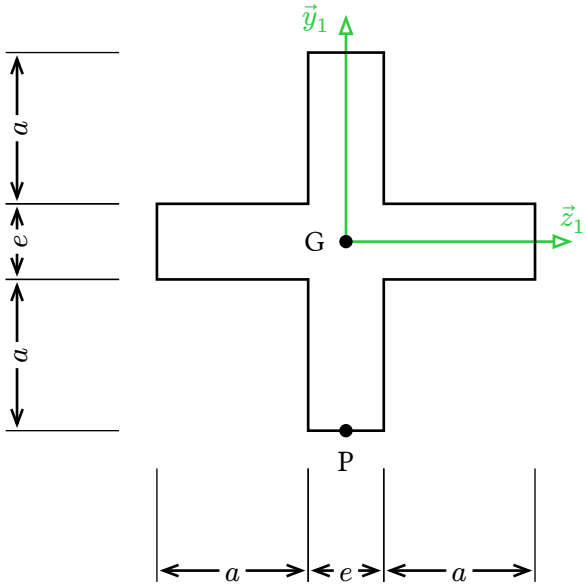
Cas du champ de vitesse d'un solide indéformable.

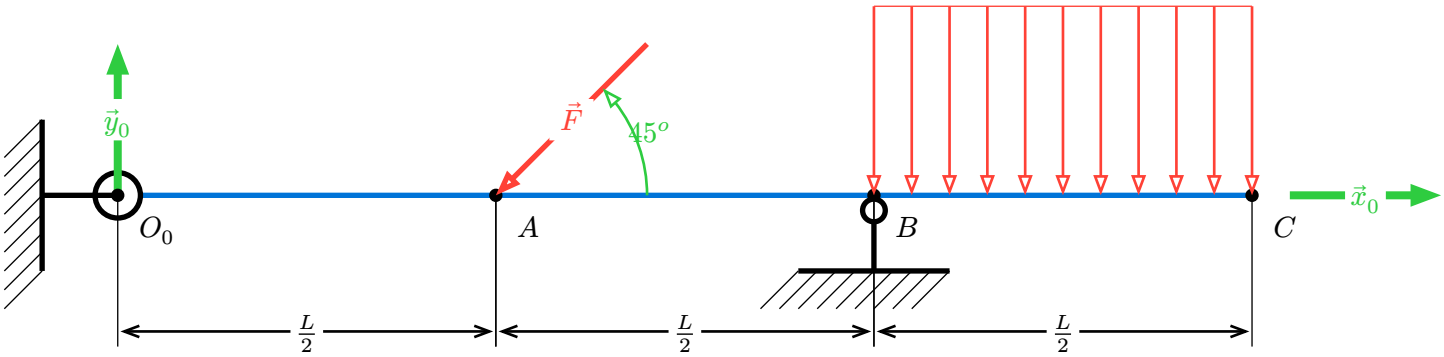



## 10 Liaisons 3D




## 11 section




## 12 Poutre